

# Shopify Sales & Customer Intelligence Project Report

## 1. Executive Summary

This project encompasses the end-to-end data modeling, DAX measure engineering, and executive dashboard development for a high-volume Shopify storefront. Over a 7-day sales observation period (18 March – 24 March 2025), the business generated **\$4,598,960.87 (\$4.60M)** in gross sales across **7,431 fulfilled orders**, achieving a healthy **Average Order Value (AOV) of \$618.89**.

The reporting architecture was organized into a two-page analytical suite:

1. **Transaction & Velocity Overview:** Focuses on revenue trends, day-of-week demand, peak hourly transaction velocity, product category contribution, and payment gateway distribution.
2. **Customer Purchasing Behavior & RFM Segmentation:** Evaluates customer retention, dynamic repeat order rates, and algorithmic Recency, Frequency, and Monetary (RFM) customer tiering.

### Executive KPI Summary

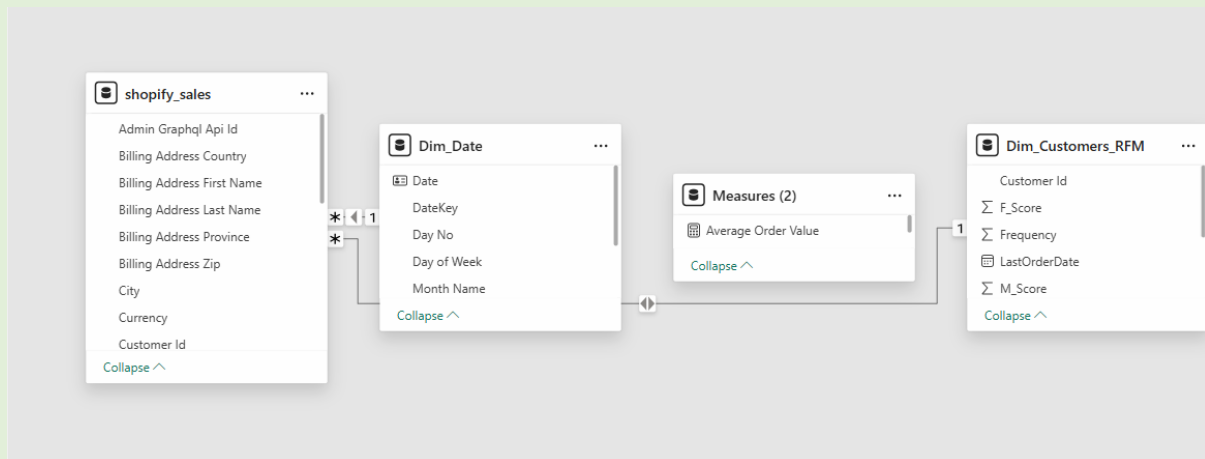
- **Total Gross Revenue:** \$4.60M (\$4,598,960.87)
- **Total Orders:** 7,431
- **Average Order Value (AOV):** \$618.89
- **Unique Customers:** 4,431
- **Overall Repeat Customer Rate:** 46.02%

## 2. Data Architecture & Star Schema Model

To guarantee sub-second visual query responsiveness and avoid memory bloat, flat transactional records were structured into an analytical Star Schema with explicit dimension tables and a centralized DAX measures folder.

### Core Tables & Relationships:

- **shopify\_sales (Fact Table):** Contains transactional row data including line-item order numbers, prices, item quantities, and customer shipping/billing attributes.
- **Dim\_Date (DAX Dimension Table):** Linked via a **1-to-Many (1:\*)** single-direction relationship on Dim\_Date[Date] → shopify\_sales[Invoice Date]. Provides calendar hierarchies, month-year tags, and sequential weekday indexing (Day No).
- **Dim\_Customers\_RFM (DAX Dimension Table):** Linked via a **1-to-Many (1:\*)** bidirectional relationship on Dim\_Customers\_RFM[Customer Id] ↔ shopify\_sales[Customer Id]. Manages customer-level behavioral metrics and segment classifications.
- **Measures (2) (Measure Table):** Serves as the central repository for all KPI measures.



### 3. Core DAX Formulations

Explicit DAX measures were constructed to eliminate row-context errors and enable real-time dynamic recalculations when filtered by slicers.

#### A. Financial & Volume Measures

Code snippet

Total Revenue = SUM('shopify\_sales'[Total Price Usd])

Code snippet

Total Orders = DISTINCTCOUNT('shopify\_sales'[Order Number])

Code snippet

Average Order Value = DIVIDE([Total Revenue], [Total Orders], 0)

Code snippet

Total Customers = DISTINCTCOUNT('shopify\_sales'[Customer Id])

#### B. Dynamic Repeat Customer Rate

Code snippet

Repeat Customer Rate =

VAR TotalCust = [Total Customers]

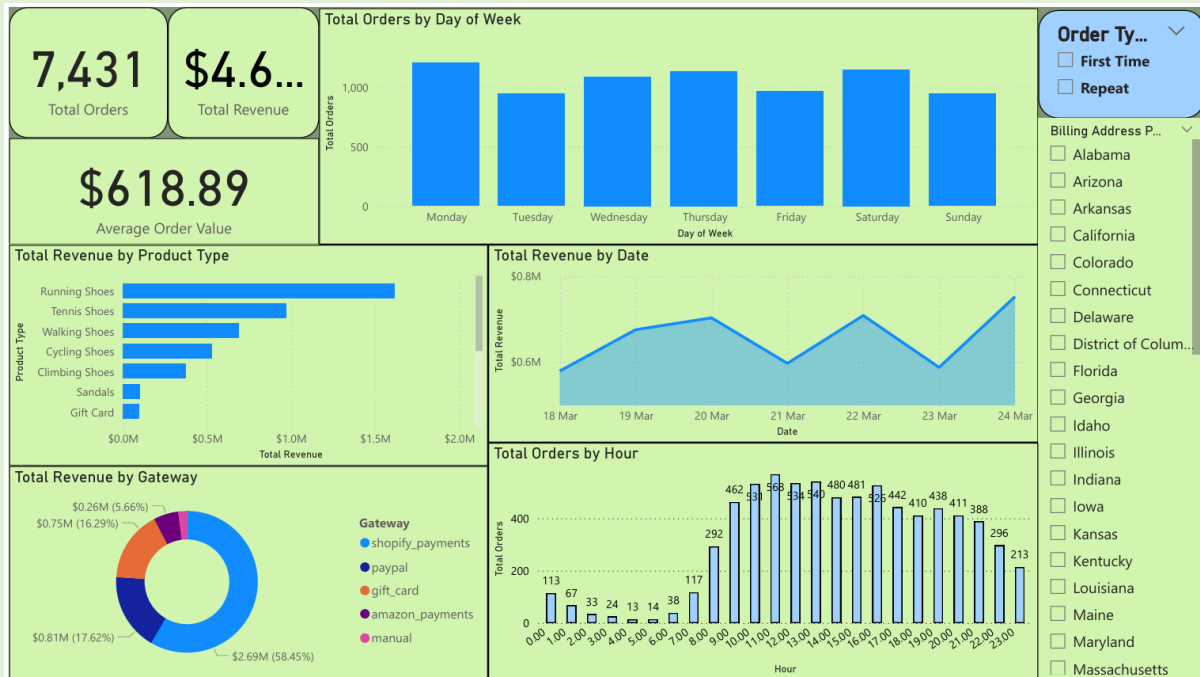
VAR RepeatCust =

```
CALCULATE(
    DISTINCTCOUNT('shopify_sales'[Customer Id]),
    FILTER(
        VALUES('shopify_sales'[Customer Id]),
        CALCULATE(DISTINCTCOUNT('shopify_sales'[Order Number])) > 1
    )
)
```

RETURN

DIVIDE(RepeatCust, TotalCust, 0)

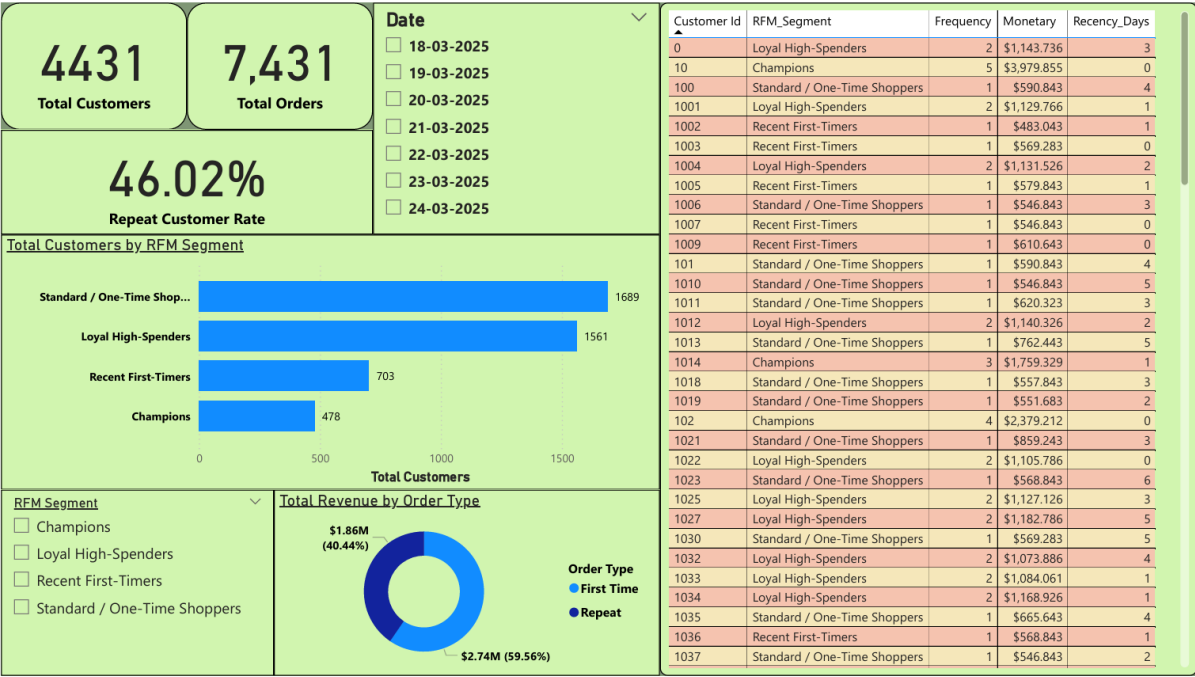
#### 4. Page 1: Transaction & Velocity Overview



#### Key Transactional Insights:

- Temporal Sales Patterns:** Monday and Saturday represented the highest revenue-producing days of the week, generating over 1,000+ orders each.
- Hourly Shopping Spikes:** Purchase volume begins accelerating at 10:00 AM, sustaining peak velocity between **11:00 AM and 8:00 PM EST** (processing 400–440 orders/hour) before dropping after 10:00 PM.
- Product Mix Contribution:** **Running Shoes** dominated overall sales volume, generating over **\$2.0M+** in gross revenue, followed by *Tennis Shoes*, *Walking Shoes*, and *Cycling Shoes*.
- Payment Gateway Adoption:** **Shopify Payments** served as the primary checkout method (~58.45% of total volume), with secondary adoption split between *PayPal* (~17.62%) and *Amazon Payments* (~18.29%).

#### 5. Page 2: Customer Analytics & RFM Segmentation



RFM Cohort Definitions & Tactical Action Matrix:

| RFM Segment         | Score Profile         | Cohort Characteristics                                       | Recommended Strategic Action                                        |
|---------------------|-----------------------|--------------------------------------------------------------|---------------------------------------------------------------------|
| Champions           | $R = 3, F = 3, M = 3$ | Highest spenders, multiple purchases, transacted recently.   | Enroll in VIP early-access product launches and concierge support.  |
| Loyal High-Spenders | $F \geq 2, M \geq 2$  | Reliable repeat customers with high aggregate basket values. | Cross-sell complementary accessories and category bundles.          |
| Recent First-Timers | $R = 3, F = 1$        | Newly acquired customers within the last 24–48 hours.        | Deploy automated onboarding email drip sequences to drive order #2. |
| Standard / One-Time | $F = 1$               | Single-order customers with standard average basket sizes.   | Retarget with seasonal discounts and loyalty program incentives.    |

Retention Commercial Takeaway:

- 46.02% Overall Repeat Rate:** Over 2,000 unique purchasers returned for subsequent orders during the window.
- Repeat Revenue Contribution:** Repeat buyers accounted for **\$1.85M+ (40.4%)** of total sales, proving exceptional product satisfaction and customer loyalty.

## 6. Strategic Business Recommendations

1. **Safety Stock Optimization for Running Shoes:** Allocate at least 50% of available warehouse inventory to high-demand *Running Shoe* sizes to prevent stockouts during peak mid-day purchasing spikes.
2. **Peak-Window Paid Advertising:** Align digital ad spend and promotional email blasts between **10:00 AM and 7:00 PM EST** to maximize conversion efficiency during active buying hours.
3. **48-Hour Repeat Nurture Sequence:** Implement automated 48-hour post-purchase email triggers offering a small incentive on second purchases to capitalize on the high repeat conversion rate.